

PROFESSIONAL SUMMARY

Computer Science graduate (BS CS, DLSU-D 2021–2025) building applied AI workflows, multi-agent systems, and full-stack web applications. Experienced in local and cloud LLM orchestration (Ollama Vulkan, Groq LPU), speech-to-speech voice DSP, real-time web intelligence, and responsive interfaces with React and TypeScript.

PROFESSIONAL WORK EXPERIENCE

UI/UX & Frontend Engineer Intern · ROC.PH Digital Marketing Services

June 2024 – August 2024 · Cavite, PH

Designed and prototyped responsive web and mobile interfaces in Figma using design token standards and usability heuristics. Translated specifications into accessible, semantic HTML5, CSS3/Tailwind, and React components across production client applications within agile sprints.

KEY TECHNICAL IMPACT

- Engineered low-latency voice pipelines with sub-350ms streaming audio DSP chunking, Whisper transcription, live meteorology, and grounded web search.
- Architected asynchronous multi-agent supervisor systems with hybrid local/cloud LLM failover (Groq LPU + AMD Vulkan) and Pydantic schema validation.
- Built lightweight episodic memory retrieval in pure Python standard library using Okapi BM25 ranking, eliminating session amnesia across agent lifecycles.
- Developed responsive, accessible web components in React, TypeScript, and modern CSS during commercial frontend internship at ROC.PH.

FEATURED ENGINEERING PROJECTS

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Voice AI Copilot & Real-Time Cognitive HUD · Solo Project

Architected conversational voice assistant with sub-350ms Thought Acceleration DSP, live meteorology (wttr.in), real-time web search (DDGS), and bidirectional Obsidian sync.
- LSFM AI HQ

Autonomous Multi-Agent Operations System · Solo Project

Architected 5 Discord agent daemons with hybrid local AMD Vulkan / Groq Cloud failover, headless Chromium ATS compiler, and OAuth2 inbox triage.
- Patriot

Community Matchmaking Mobile App · DLSU-D Capstone (Lead Mobile Developer)

Engineered real-time mobile app in React Native/TypeScript with AWS Lambda, DynamoDB, Redis caching for session state, and multi-criteria matching.
- Cognitive Memory Core

Episodic Memory Retrieval Engine · Solo Project

Engineered zero-dependency episodic memory retrieval in Python standard library using BM25 keyword relevance scoring for pre-flight task context.

EDUCATION

Bachelor of Science in Computer Science (BS CS)

2021 – 2025

De La Salle University–Dasmariñas (DLSU-D) Distributed Systems · Human-Computer Interaction · Algorithms & Complexities

TECHNICAL PROFICIENCIES & CAPABILITIES

AI & LLM Orchestration:	Multi-Agent Workflows, Hybrid Vector RAG (Dense + BM25 with RRF), Voice AI DSP, Local GPU Inference (Ollama, Vulkan), Cloud LLMs (Groq, Google Gemini, OpenAI), Tool Calling, Structured JSON
Languages & Backend:	Python 3.11 (asyncio, OOP, standard library), TypeScript, JavaScript (ES6+), Node.js, Express, aiohttp, REST APIs, WebSocket, OAuth2, Headless Chromium, Git/GitHub
Frontend & Design Systems:	React.js, React Native, HTML5, Modern CSS3, Design Tokens, Web Audio API, Canvas 2D, Tailwind CSS, Responsive Web Design, WCAG 2.1 Accessibility, Figma